

ISLE OF MAN, Isle of Man

WMO: 032040

Lat: 54.0853N

Lon: 4.6319W

Elev: 55

StdP: 14.67

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.6	33.6	22.0	16.6	36.1	24.1	18.3	36.6	41.6	46.1	37.7	44.9	9.7	10	0.739

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	69.7	60.9	66.5	59.6	64.4	58.5	62.4	67.0	61.0	64.3	60.0	62.9	10.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	78.8	63.5	59.5	76.1	62.4	58.7	73.8	61.6	28.0	67.5	27.1	64.5	26.4	62.9	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.5	31.7	28.5	DB	27.8	74.4	2.4	3.8	26.0	77.2	24.6	79.4	23.2	81.6	21.5	84.4	(4)
(5)			WB	25.8	65.0	2.5	1.9	24.0	66.3	22.6	67.5	21.2	68.5	19.4	69.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	43.4	42.8	44.0	47.0	51.6	55.8	59.1	59.3	56.8	52.8	47.9	44.6	(6)
(7)		DBStd	7.08	4.26	4.04	3.78	3.63	3.95	3.36	3.11	2.62	2.83	3.84	4.49	4.87	(7)
(8)		HDD50	998	205	202	186	101	25	1	0	0	1	19	87	171	(8)
(9)		HDD65	5307	669	621	650	540	415	276	188	180	246	379	512	631	(9)
(10)		CDD50	1172	1	0	1	11	75	175	281	288	205	104	26	5	(10)
(11)		CDD65	7	0	0	0	0	0	1	4	2	0	0	0	0	(11)
(12)		CDH74	21	0	0	0	0	0	4	12	5	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	14.1	17.6	16.6	15.0	12.8	12.2	11.2	10.5	11.3	12.8	15.6	16.1	17.4	(14)
(15)	Precipitation	PrecAvg	37.6	3.7	2.9	2.6	2.4	2.4	2.6	2.5	3.0	4.4	4.4	4.2	(15)	
(16)		PrecMax	49.8	7.6	5.4	7.1	5.3	5.4	7.7	3.9	6.1	6.4	10.0	9.1	8.2	(16)
(17)		PrecMin	28.5	0.6	0.7	0.9	0.6	0.6	0.7	1.1	0.3	0.2	1.6	2.3	2.3	(17)
(18)		PrecStd	5.3	1.6	1.3	1.3	1.1	1.3	1.5	0.8	1.4	1.4	2.1	1.7	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	51.9	53.9	61.2	69.1	71.9	75.8	72.9	66.4	61.3	57.6	54.8	(19)
(20)			MCWB	50.6	49.1	48.5	52.1	57.6	59.9	63.5	63.0	60.6	57.5	55.6	52.8	(20)
(21)		2%	DB	51.5	50.4	51.8	56.6	63.9	67.1	70.4	67.9	63.8	60.0	56.3	53.6	(21)
(22)			MCWB	49.5	48.2	47.9	49.6	55.9	58.9	61.8	61.2	58.9	57.0	54.2	51.7	(22)
(23)		5%	DB	50.4	49.5	50.3	53.9	60.4	63.7	66.5	65.3	62.4	59.1	55.2	52.5	(23)
(24)			MCWB	48.4	47.4	47.1	48.5	54.0	56.9	59.7	59.6	58.0	56.5	53.0	50.6	(24)
(25)		10%	DB	49.5	48.5	49.2	52.2	57.6	61.3	64.0	63.5	61.2	58.1	54.2	51.4	(25)
(26)			MCWB	47.4	46.4	46.4	47.9	52.3	55.7	58.5	58.6	57.5	55.5	51.8	49.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.4	50.3	50.4	52.9	59.4	62.2	64.8	64.3	62.0	59.2	56.4	53.6	(27)
(28)			MADB	52.2	51.0	52.1	58.9	66.3	68.6	72.5	70.4	64.5	60.1	57.1	54.1	(28)
(29)		2%	WB	50.2	49.1	49.2	51.0	56.7	59.7	62.6	62.2	60.5	58.2	54.8	52.4	(29)
(30)			MADB	51.1	50.0	50.6	54.7	62.5	65.2	68.5	65.9	62.1	59.4	55.9	53.3	(30)
(31)		5%	WB	49.1	48.1	48.1	49.8	54.9	58.1	61.0	61.0	59.6	57.2	53.6	51.0	(31)
(32)			MADB	50.3	49.2	49.5	52.7	59.2	62.5	65.2	63.6	61.4	58.7	54.9	52.3	(32)
(33)		10%	WB	47.9	46.9	47.1	48.8	53.2	56.7	59.7	60.1	58.5	56.1	52.3	49.7	(33)
(34)			MADB	49.3	48.4	48.7	51.4	56.7	60.2	63.0	62.4	60.6	57.9	54.0	51.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	6.8	7.7	8.8	9.6	9.2	9.0	8.7	8.1	7.2	6.8	6.7	(35)
(36)			MCDBR	6.6	6.2	7.9	11.0	13.7	13.2	13.2	11.0	9.0	7.1	6.4	6.9	(36)
(37)			MCWBR	7.6	6.3	6.3	7.1	8.3	7.6	7.4	6.9	6.6	6.6	7.5	7.8	(37)
(38)		5% WB	MCDBR	6.6	6.0	7.1	8.9	11.9	11.3	11.0	9.0	7.8	6.3	6.3	6.6	(38)
(39)			MCWBR	7.8	6.6	6.6	6.5	7.9	7.2	7.0	6.4	6.7	6.8	7.6	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.333	0.344	0.370	0.388	0.392	0.386	0.387	0.373	0.366	0.351	0.328	0.320	(40)
(41)		taud		2.292	2.372	2.341	2.328	2.374	2.423	2.421	2.474	2.488	2.457	2.365	2.268	(41)
(42)		Ebn at Noon		192	236	255	266	271	274	271	269	256	232	195	170	(42)
(43)		Edh at Noon		19	24	31	36	36	34	34	30	27	22	18	16	(43)
(44)	All-Sky Solar Radiation	RadAvg		187	408	794	1272	1628	1676	1571	1307	914	502	238	138	(44)
(45)		RadStd		18	38	60	89	127	151	129	93	57	49	24	13	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.32	+0.77	+1.35	N/A	N/A	N/A	N/A	-135	+81	N/A	
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page